Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years building full-stack, database-backed business applications and cross-platform services. Experienced in Java, Python, C++, SQL Server, Azure SQL, REST APIs, object-oriented design, testing, troubleshooting, and end-user workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected Azure SQL and Blob Storage services for 500+ users, refining normalized schemas, indexed queries, and secure access patterns to reduce member-facing dashboard latency from 1.2 seconds to 350 milliseconds.
- Refactored fault-tolerant background workflows for 1,000+ jobs using explicit states, retries, and failure isolation, tripling concurrent throughput while improving maintainability, operational observability, and reliable production support.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed responsive React components and Flask REST services for secure upload, routing, status, and history workflows, translating user requirements into maintainable features supporting 150+ users and 10,000+ daily requests.
- Created concurrent processing for 1,000+ files with automated validation, error handling, tests, and AWS RDS persistence, reducing turnaround 60% while maintaining stable application behavior during peak workloads.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented API and ETL components using reusable software design patterns, AWS Glue, and S3, improving response times 30% and pipeline efficiency 68% across complex customer workflows.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving engineering confidence in production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular MERN services and WebSocket synchronization across eight customer, billing, scheduling, permissions, and notification workflows, supporting 2 million+ Azure SQL records with independently maintainable service boundaries.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving overall software quality and consistent operational reliability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed the end-to-end application workflow, mapped 12+ backend integration points, and presented technical architecture to a client before implementing cross-platform booking and payment experiences handling 50,000+ records.
- Developed Flask APIs for bookings, pricing, and payments with sub-200-millisecond responses, transactional consistency, and webhook automation, reducing manual payment handling 70% while improving customer confirmation speed 40%.

Skills

Languages & Application Development: Java, Python, C++, TypeScript, JavaScript, SQL, Bash, object-oriented programming, design patterns, full-stack development, desktop support
Backend & Business Systems: REST APIs, Flask, FastAPI, Node.js, microservices, WebSockets, requirements gathering, end-user workflows, API integration, troubleshooting, implementation
Databases & Cloud: SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, AWS RDS, Azure, AWS, data modeling, relational databases, indexing, query optimization, transactions
Quality & Delivery: Git, Jenkins, CI/CD, Docker, PyTest, Mockito, unit testing, integration testing, automated testing, debugging, code review, documentation, Agile, Scrum, production support

Projects

FinNexus | Live | *TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a financial platform supporting 50,000+ users and one million transactions, integrating secure bank APIs, real-time synchronization, advanced filtering, database-backed workflows, and reliable containerized production deployment.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a cross-platform SQL development workspace supporting 10+ database types, combining Java services, schema-aware query generation, validation, Docker, and Maven workflows across Windows, macOS, and Linux systems.

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time 20% through deterministic simulation, maintainable C++ logic, and systematic edge-case testing.